INTELLIGENT AGENTS

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Agents in AI

- *An **AI agent** is a software program that can interact with its surroundings, gather information, and use that information to complete tasks on its own to achieve goals set by humans*
- For instance, an **AI agent** on an online shopping platform can recommend products, answer customer questions, and process orders. If agent needs more information, it can ask users for additional details.
- AI agents employ advanced **natural language processing** and **machine learning** techniques to understand user input, interact step-by-step, and use external tools when needed for accurate responses.
- Common AI Agent Applications are *software development and IT automation, coding tools, chat assistants, and online shopping platforms*

How do AI Agents Work?

1. Collecting Information (Perceiving the Environment)

- AI agents gather information from their surroundings through various means:
- **Sensors:** For example, a self-driving car uses cameras and radar to detect objects.
- **User Input:** Chatbots read text or listen to voice commands.
- **Databases & Documents:** Virtual assistants search records or knowledge bases for relevant data.

2. Processing Information & Making Decisions

After gathering data, AI agents analyze it and decide what to do next. Some agents rely on **pre-set rules**, while others utilize **machine learning** to predict the best course of action. Advanced agents may also use **retrieval-augmented generation (RAG)** to access external databases for more accurate responses.

3. Taking Action (Performing Tasks)

Once an agent makes a decision, it performs the required task, such as:

- Answering a customer query in a chatbot.
- Controlling a device, like a smart assistant turning off lights.
- Running automated tasks, such as processing orders on an online store.

4. Learning & Improving Over Time

Some AI agents can **learn** from past experiences to improve their responses. This self-learning process, often referred to as **reinforcement learning**, allows agents to refine their behavior over time. For example, a **recommendation system** on a streaming platform learns users' preferences and suggests content accordingly.

AI Agent Classification

In artificial intelligence (AI), reactive agents respond to immediate stimuli, while proactive agents plan ahead to achieve goals:

- **Reactive agents**

These agents perform simple tasks by retrieving pre-set behaviors. They don't maintain an internal state, but instead act based on what they perceive in their environment. Examples of reactive AI include spam filters and Netflix's recommendation engine. Reactive agents are also used in robotics for tasks like obstacle avoidance and navigation.

- **Proactive agents**

These agents take initiative and plan ahead to achieve their goals. They can analyze data, predict outcomes, and act accordingly. Proactive agents are model-based reflex agents that maintain an internal state and evaluate probable outcomes based on an internal model of the world. Examples of proactive AI include autonomous vehicles and proactive AI chatbots.

The **environment** in which an agent operates can also be **fixed** or **dynamic**. Fixed environments have a static set of rules that do not change, while dynamic environments are constantly changing and require agents to adapt to new situations.

In artificial intelligence (AI), the main difference between single-agent and multi-agent is how they work:

- **Single-agent systems --** These systems are best for well-defined tasks that don't require interaction with other agents. They are easier to develop and deploy, and they have centralized control. However, they can be rigid and struggle in complex or rapidly changing environments.
- **Multi-agent systems --** These systems are best for complex operations that benefit from specialized agents working together. They are more adaptable and scalable than single-agent systems, and they can outperform them in terms of accuracy and efficiency.

Examples -- Solitaire is an example of a single-agent environment, while a soccer match is an example of a multi-agent environment.

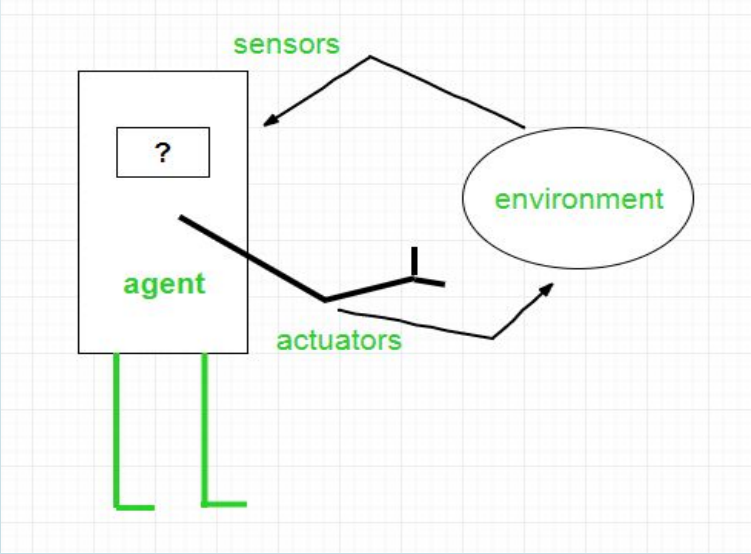
Use cases -- Single-agent systems can be used for customer support, data retrieval, and task automation. Multi-agent systems can be used to integrate agents with expertise in marketing, finance, supply chain management, or customer service

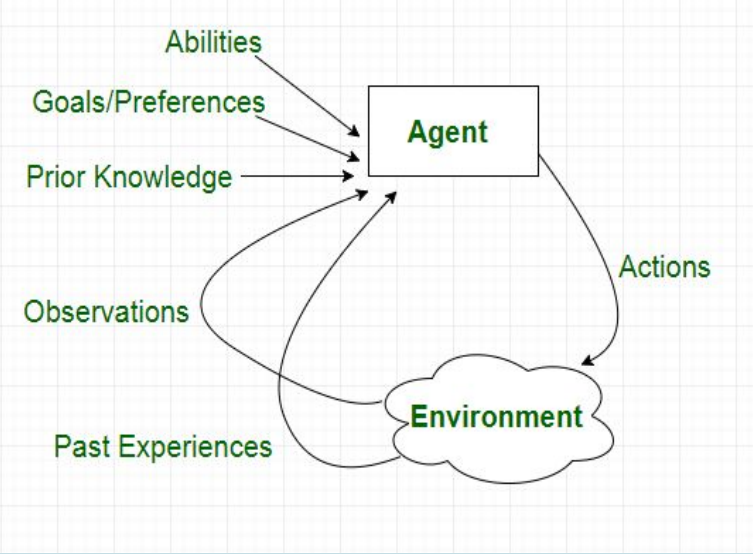
- **A rational agent could** be anything that makes decisions, such as a person, firm, machine, or software. It carries out an action with the best outcome after considering past and current percepts (agent's perceptual inputs at a given instance). An AI system is composed of an agent and its environment. The agents act in their environment. The environment may contain other agents.
- Perceiving its environment through sensors and
- Acting upon that environment through actuators

Key Components of an AI System

- An AI system includes the **agent**, which perceives the environment through sensors and acts using actuators, and the **environment**, in which it operates.
- AI agents are essential in fields like robotics, gaming, and intelligent systems, where they use various techniques such as machine learning to enhance decision-making and adaptability

Agent = Architecture + Agent Program



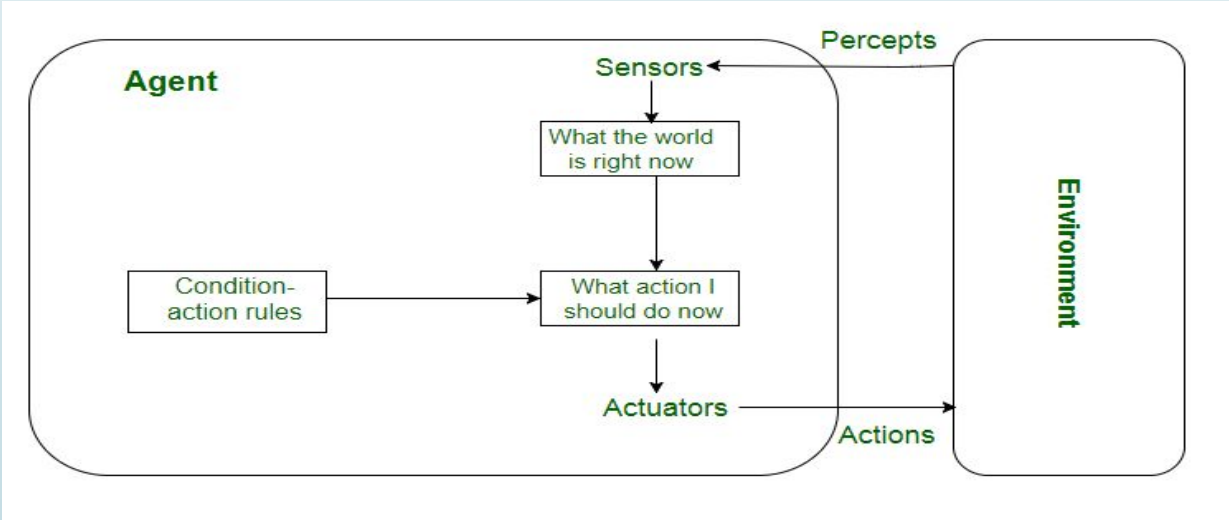


Types of Agents
1. Simple Reflex Agents

- Simple reflex agents act solely based on the **current percept and**, **percept history** (record of past perceptions) is ignored by these agents. Agent function is defined by **condition-action rules**.

A **condition-action rule** maps a state (condition) to an action.

- If the condition is true, the associated action is performed.
- If the condition is false, no action is taken.
- Simple reflex agents are effective in environments that are fully observable (where the current percept gives all needed information about the environment). In **partially observable environments**, simple reflex agents may encounter **infinite loops** because they do not consider the history of previous percepts.



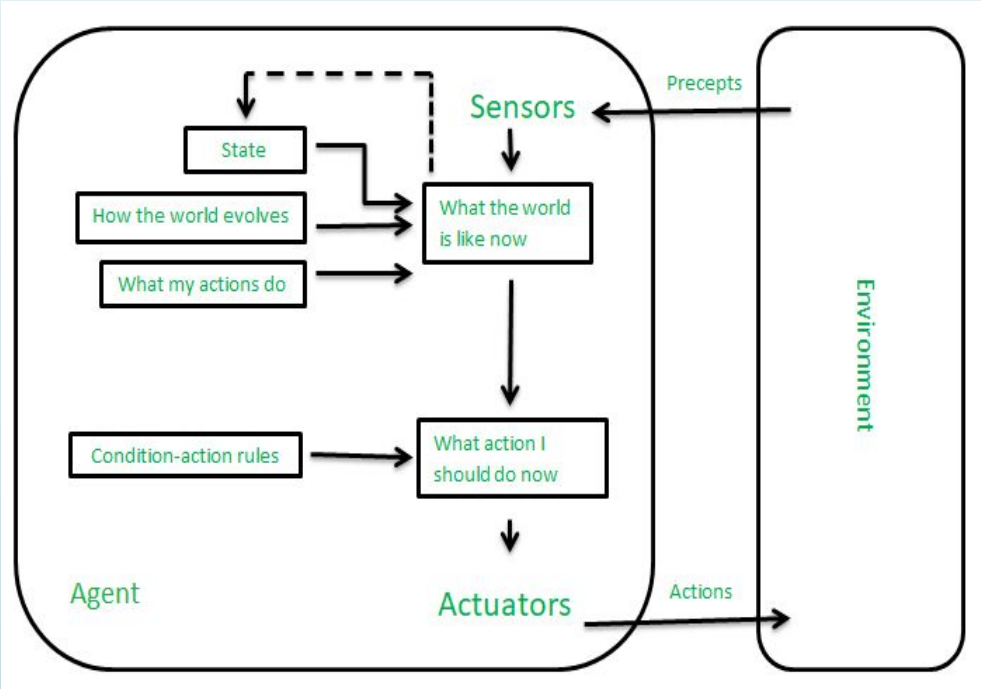
2. Model-Based Reflex Agents

Model-based reflex agents finds a rule whose condition matches the current situation or percept. It uses a model of the world to handle situations where the environment is only partially observable.

- The agent tracks its internal state, which is adjusted based on each new percept.
- The internal state depends on the percept history (the history of what the agent has perceived so far).

The agent stores the current state internally, maintaining a structure that represents the parts of the world that cannot be directly seen or perceived. The process of updating the agent's state requires information about:

- How the world evolves independently from the agent?
- How the agent's actions affect the world?

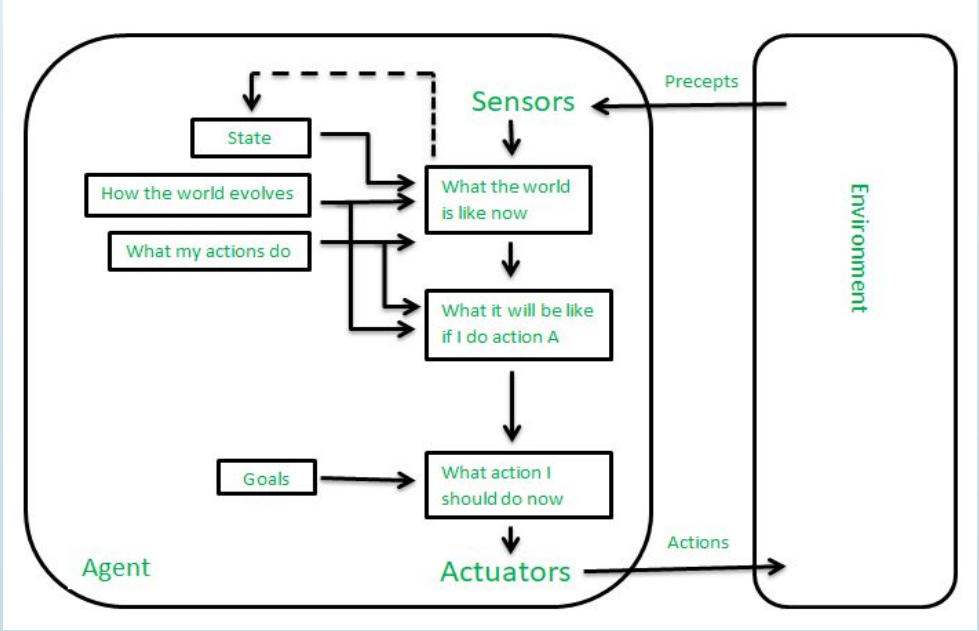


3. Goal-Based Agents

Goal-based agents make decisions based on their current distance from the goal and every action the agent aims to reduce the distance from goal. They can choose from multiple possibilities, selecting the one that best leads to the goal state

- Knowledge that supports the agent's decisions is represented explicitly, meaning it's clear and structured. It can also be modified, allowing for adaptability.
- The ability to modify the knowledge makes these agents more flexible in different environments or situations.

Goal-based agents typically require search and planning to determine the best course of action.

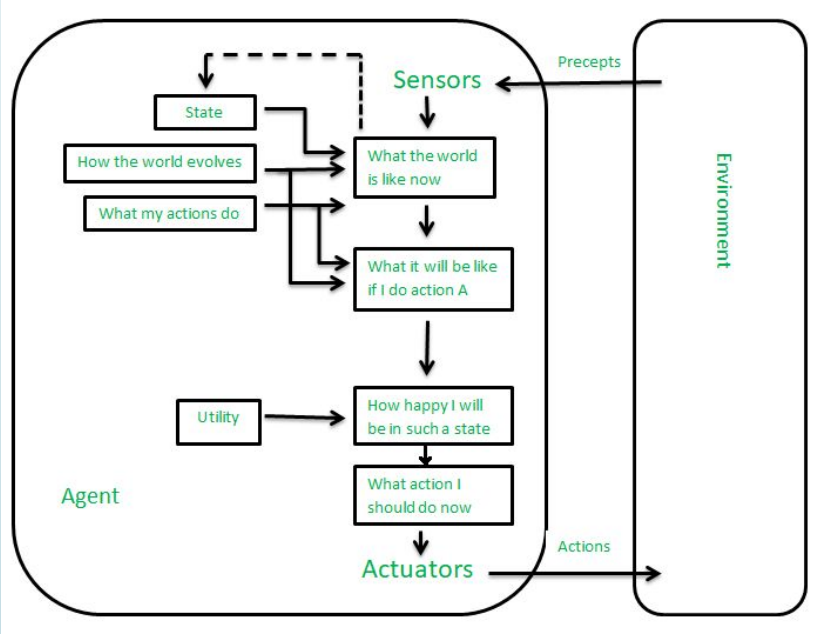


4. Utility-Based Agents

Utility-based agents are designed to make decisions that optimize their performance by evaluating the preferences (or utilities) for each possible state. These agents assess multiple alternatives and choose the one that maximizes their utility, which is a measure of how desirable or "happy" a state is for the agent.

- Achieving the goal is not always sufficient; for example, the agent might prefer a quicker, safer, or cheaper way to reach a destination.
- The utility function is essential for capturing this concept, mapping each state to a real number that reflects the agent's happiness or satisfaction with that state.

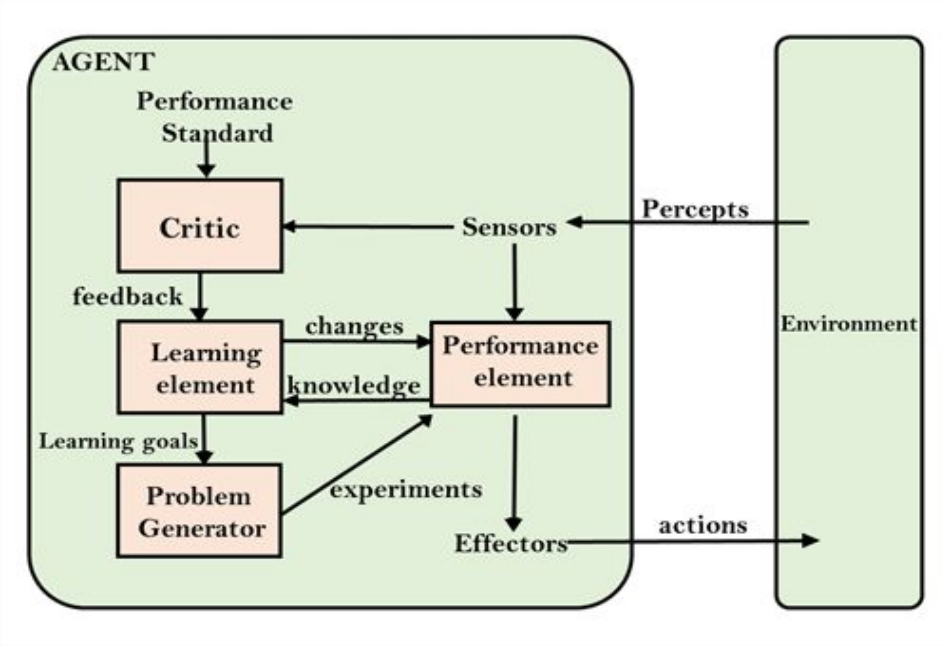
Since the world is often uncertain, utility-based agents choose actions that maximize expected utility, ensuring they make the most favorable decision under uncertain conditions.



5. Learning Agent

A learning agent in AI is the type of agent that can learn from its past experiences or it has learning capabilities. It starts to act with basic knowledge and then is able to act and adapt automatically through learning. A learning agent has mainly four conceptual components, which are:

- Learning element: It is responsible for making improvements by learning from the environment.
- Critic: The learning element takes feedback from critics which describes how well the agent is doing with respect to a fixed performance standard.
- Performance element: It is responsible for selecting external action.
- Problem Generator: This component is responsible for suggesting actions that will lead to new and informative experiences.



Uses of Agents

- **Robotics:** Agents can be used to control robots and automate tasks in manufacturing, transportation, and other industries.
- **Smart homes and buildings:** Agents can be used to control heating, lighting, and other systems in smart homes and buildings, optimizing energy use and improving comfort.
- **Transportation systems:** Agents can be used to manage traffic flow, optimize routes for autonomous vehicles, and improve logistics and supply chain management.
- **Healthcare:** Agents can be used to monitor patients, provide personalized treatment plans, and optimize healthcare resource allocation.
- **Finance:** Agents can be used for automated trading, fraud detection, and risk management in the financial industry.
- **Games:** Agents can be used to create intelligent opponents in games and simulations, providing a more challenging and realistic experience for players.